

A TREE BANACH-SAKS PARTIAL ANSWER TO BRAGA'S WEAKLY SEQUENTIALLY CONTINUOUS COARSE EMBEDDING PROBLEM

AGENT LANE 19

1. SOURCE PROBLEM

The source is Bruno de Mendonca Braga's paper *On asymptotically uniformly smoothness and nonlinear geometry of Banach spaces*, arXiv:1808.03254 [1]. In Section 8, Problem 8.6 asks:

Let X and Y be Banach spaces. Assume that Y has the weak p -Banach-Saks property, for some $p \in (1, \infty)$. If X coarse Lipschitzly embeds into Y by a weakly sequentially continuous map, does it follow that X has the weak p -Banach-Saks property?

The crop below shows the source question on page 33 of the compiled source PDF.

coarse Lipschitz embedding from X into Y ?

At last, we ask whether a version of Theorem 6.10 holds for the weak p -Banach-Saks property. Precisely, we ask the following.

Problem 8.6. Let X and Y be Banach spaces. Assume that Y has the weak p -Banach-Saks property, for some $p \in (1, \infty)$. If X coarse Lipschitzly embeds into Y by a weakly sequentially continuous map, does it follow that X has the weak p -Banach-Saks property?

2. DEFINITIONS AND STATUS

A weakly null tree $(x_{\bar{n}})_{\bar{n} \in [\mathbb{N}]^{\leq k}}$ is indexed by increasing finite sequences. A Banach space X has the tree- p -Banach-Saks property with constant C if every normalized weakly null tree of height k in X has a branch $n_1 < \dots < n_k$ such that

$$\left\| \sum_{j=1}^k x_{n_1, \dots, n_j} \right\| \leq Ck^{1/p}.$$

Braga records that tree- p -Banach-Saks implies weak p -Banach-Saks [1, Section 6].

This packet proves the permanence statement asked in Problem 8.6 under the stronger hypothesis that Y has tree- p -Banach-Saks. Thus the original weak p -Banach-Saks target hypothesis remains open.

3. PROOF INTUITION

Braga's Theorem 6.10 proves a related permanence theorem by applying a weakly sequentially continuous coarse Lipschitz embedding to partial sums along a weakly null tree. The successive differences in Y form another weakly null tree, and the telescoping sum of those differences is exactly the embedded image of the original branch sum.

The only extra point here is that tree- p -Banach-Saks controls equal-coefficient sums, while the telescoping differences have different lengths. The sign-stabilized form of tree- p -Banach-Saks gives one branch on which all signed sums of the normalized differences are small. Any positive coefficient vector with coefficients bounded by 1 is an average of sign vectors, so the same branch also controls the actual variable-length telescoping sum.

4. THE PARTIAL THEOREM

Theorem 1. *Let $p \in (1, \infty)$, and let X and Y be Banach spaces. Assume that Y has the tree- p -Banach-Saks property. If X coarse Lipschitzly embeds into Y by a weakly sequentially continuous map, then X has the tree- p -Banach-Saks property. Consequently, X has the weak p -Banach-Saks property.*

Proof. Let C witness the tree- p -Banach-Saks property of Y . We first use the standard sign form of this property: for every normalized weakly null tree $(z_{\bar{n}})_{\bar{n} \in [\mathbb{N}]^{\leq k}}$ in Y , there is an infinite $M \subset \mathbb{N}$ such that

$$\left\| \sum_{j=1}^k \varepsilon_j z_{n_1, \dots, n_j} \right\| \leq Ck^{1/p}$$

for all $n_1 < \dots < n_k$ in M and all signs $\varepsilon_j \in \{-1, 1\}$. This is Braga's Proposition 6.5, obtained by applying the tree property to the finitely many sign-changed trees and then using Ramsey stabilization.

Let $f : X \rightarrow Y$ be a weakly sequentially continuous coarse Lipschitz embedding. After translating, assume $f(0) = 0$. There are constants $L \geq 1$ and $A \geq 0$ such that

$$L^{-1}\|x - y\| - A \leq \|f(x) - f(y)\| \leq L\|x - y\| + A$$

for all $x, y \in X$. For $m \in \mathbb{N}$, define $f_m(x) = m^{-1}f(mx)$. Then f_m is still weakly sequentially continuous and satisfies the same inequalities with A/m in place of A . Choose m so large that $\delta = A/m < L^{-1}/2$. Put $U = L + \delta$.

Fix $k \in \mathbb{N}$ and let $(x_{\bar{n}})_{\bar{n} \in [\mathbb{N}]^{\leq k}}$ be a normalized weakly null tree in X . For a node $s = (n_1, \dots, n_j)$, write

$$u_s = \sum_{i=1}^j x_{n_1, \dots, n_i}, \quad u_{\emptyset} = 0,$$

and let $s^- = (n_1, \dots, n_{j-1})$ be the predecessor of s . Define a tree in Y by

$$y_s = f_m(u_s) - f_m(u_{s^-}).$$

For each fixed predecessor s , the sequence $x_{s,n}$ is weakly null, hence $u_s + x_{s,n}$ converges weakly to u_s . Since f_m is weakly sequentially continuous, $f_m(u_s + x_{s,n}) - f_m(u_s)$ is weakly null. Therefore (y_s) is a weakly null tree.

Moreover, because every x_s is normalized,

$$L^{-1} - \delta \leq \|y_s\| \leq L + \delta = U$$

for every node s . Thus $e_s = y_s / \|y_s\|$ is a normalized weakly null tree in Y . Applying the sign form above, choose a branch $n_1 < \dots < n_k$ such that all signed sums of $e_{n_1, \dots, n_j}$ are bounded by $Ck^{1/p}$. For this branch, set

$$t_j = \frac{\|y_{n_1, \dots, n_j}\|}{U} \in [0, 1].$$

The vector $\sum_{j=1}^k t_j e_{n_1, \dots, n_j}$ lies in the convex hull of the sign sums $\sum_{j=1}^k \varepsilon_j e_{n_1, \dots, n_j}$: choose independent signs with mean t_j and take expectations. Hence

$$\left\| \sum_{j=1}^k y_{n_1, \dots, n_j} \right\| = U \left\| \sum_{j=1}^k t_j e_{n_1, \dots, n_j} \right\| \leq UCk^{1/p}.$$

The left side telescopes to $\|f_m(u_{n_1, \dots, n_k}) - f_m(0)\|$. The lower coarse Lipschitz inequality gives

$$L^{-1} \|u_{n_1, \dots, n_k}\| - \delta \leq UCk^{1/p}.$$

Since $k^{1/p} \geq 1$,

$$\left\| \sum_{j=1}^k x_{n_1, \dots, n_j} \right\| \leq L(UC + \delta)k^{1/p}.$$

This proves that X has the tree- p -Banach-Saks property. Braga's observation that tree- p -Banach-Saks implies weak p -Banach-Saks then gives the final conclusion. $\square$

5. VERIFICATION AND LIMITATIONS

The proof is purely analytic. The main verification points are:

- weak sequential continuity is used exactly to turn the nonlinear increments y_s into a weakly null tree;
- rescaling f makes the additive coarse error small enough that those increments are semi-normalized;
- the sign-stabilized tree- p -Banach-Saks branch plus convexity handles the variable increment lengths;
- the final telescoping identity transfers the branch estimate back to X .

The result does not settle Braga’s Problem 8.6 as stated. It replaces the target assumption “ Y has weak p -Banach–Saks” by the stronger assumption “ Y has tree- p -Banach–Saks.” The endpoint p is preserved.

Cheap run indexes were searched for ‘1808.03254’, Braga, weak p -Banach–Saks, tree-Banach–Saks, and weakly sequentially continuous coarse Lipschitz embeddings. Web searches for the exact problem label and core phrase combinations found the source arXiv page and no later direct answer. A later local source on Hamming graph concentration properties was checked for overlap and did not answer Problem 8.6.

6. REFERENCES

REFERENCES

- [1] B. de Mendonca Braga, *On asymptotically uniform smoothness and nonlinear geometry of Banach spaces*, arXiv:1808.03254, 2018. <https://arxiv.org/abs/1808.03254>